CONTROL SYSTEM

Unit	Content	Hrs./ Week	Marks
1	**Introduction to Control Systems** 1.1 Definition and Need for Control Systems 1.2 Open-loop and Closed-loop Control Systems 1.3 Feedback Mechanism and Its Importance 1.4 Examples of Control Systems in Industrial Automation	6	8
2	**Mathematical Modeling of Control Systems** 2.1 Transfer Function and System Representation 2.2 Differential Equations for Electrical, Mechanical, and Hydraulic Systems 2.3 Poles, Zeros, and System Order 2.4 Block Diagram Representation and Reduction Techniques 2.5 Signal Flow Graphs and Mason's Gain Formula	6	10
3	**Time-Domain Analysis of Control Systems** 3.1 Time Response of First and Second Order Systems 3.2 Standard Test Signals (Step, Ramp, Impulse, Parabolic) 3.3 Time-domain Specifications (Rise Time, Settling Time, Peak Time, etc.) 3.4 Stability and Types of Stability 3.5 Routh-Hurwitz Stability Criterion	6	12
4	**Frequency-Domain Analysis of Control Systems** 4.1 Introduction to Frequency Response 4.2 Bode Plot: Construction and Interpretation 4.3 Nyquist Plot and Stability Criterion 4.4 Gain and Phase Margins	6	12
5	**Control System Components and Controllers** 5.1 Introduction to Actuators and Sensors 5.2 Types of Controllers: P, PI, PD, PID Controllers 5.3 Comparison of Different Controllers 5.4 Tuning of PID Controllers 5.5 Industrial Applications of PID Controllers	6	12
6	**State-Space Analysis of Control Systems** 6.1 State Variables and State Equations 6.2 State Transition Matrix 6.3 Solution of State Equations 6.4 Controllability and Observability	6	12
7	**Stability Analysis Using Root Locus** 7.1 Root Locus Concept and Rules 7.2 Construction of Root Locus Diagram 7.3 Effect of Pole and Zero Locations on Stability	6	12

Unit	Content	Hrs./ Week	Marks
8	**Modern Control Systems and Applications** 8.1 Digital Control Systems and Introduction to PLC-Based Control 8.2 Introduction to Fuzzy Logic and Neural Network Control 8.3 Industrial Applications of Control Systems (Robotics, Automation, Process Control)	6	12
	TOTAL	**48**	**100**